

torch.func.grad_and_value#

https://pytorch.org/docs/stable/generated/torch.func.grad_and_value.html

Description:

Returns a function to compute a tuple of the gradient and primal, or forward, computation. `func` (Callable) – A Python function that takes one or more arguments. Must return a single-element Tensor. If specified `has_aux` equals `True`, function can return a tuple of single-element Tensor and other auxiliary objects: `(output, aux)`. `argnums` (int or `Tuple[int]`) – Specifies arguments to compute gradients with respect to. `argnums` can be single integer or tuple of integers. Default: 0. `has_aux` (bool) – Flag indicating that `func` returns a tensor and other auxiliary objects: `(output, aux)`. Default: `False`. Function to compute a tuple of gradients with respect to its inputs and the forward computation. By default, the output of the function is a tuple of the gradient tensor(s) with respect to the first argument and the primal computation. If specified `has_aux` equals `True`, tuple of gradients and tuple of the forward computation with output auxiliary objects is returned. If `argnums` is a tuple of integers, a tuple of a tuple of the output gradients with respect to each `argnums` value and the forward computation is returned.

Function Signature

```
torch.func.grad_and_value(func, argnums=0, has_aux=False)  
[source]#
```

Parameters

Returns

Return type

Returns:

Callable

Detailed Documentation

Documentation

Returns a function to compute a tuple of the gradient and primal, or forward, computation.

`func` (Callable) – A Python function that takes one or more arguments. Must return a single-element Tensor. If specified `has_aux` equals `True`, function can return a tuple of single-element Tensor and other auxiliary objects: (output, aux).

`argnums` (int or Tuple[int]) – Specifies arguments to compute gradients with respect to. `argnums` can be single integer or tuple of integers. Default: 0.

`has_aux` (bool) – Flag indicating that `func` returns a tensor and other auxiliary objects: (output, aux). Default: `False`.

Function to compute a tuple of gradients with respect to its inputs and the forward computation. By default, the output of the function is a tuple of the gradient tensor(s) with respect to the first argument and the primal computation. If specified `has_aux` equals `True`, tuple of gradients and tuple of the forward computation with output auxiliary objects is returned. If `argnums` is a tuple of integers, a tuple of a tuple of the output gradients with respect to each `argnums` value and the forward computation is returned.

See `grad()` for examples